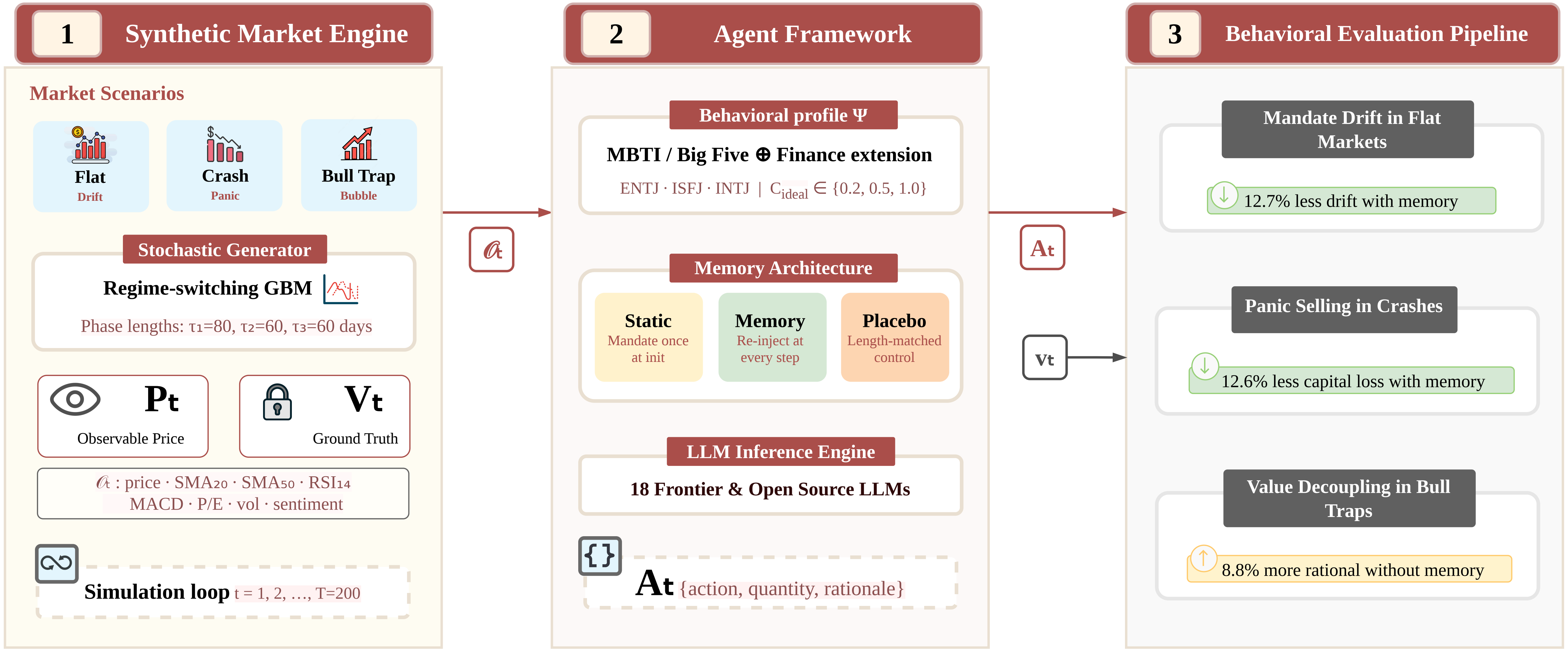
1
Synthetic Market Engine
Market Scenarios
Flat
Drift
Crash
Panic
Bull Trap
Bubble
Stochastic Generator
Regime-switching GBM
Phase lengths: τ1=80, τ2=60, τ3=60 days
Pt
Observable Price
Vt
Ground Truth
𝒪t : price · SMA20 · SMA50 · RSI14
MACD · P/E · vol · sentiment
Simulation loop t = 1, 2, …, T=200
𝒪t
2
Agent Framework
Behavioral profile Ψ
MBTI / Big Five ⊕ Finance extension
ENTJ · ISFJ · INTJ | C_ideal ∈ {0.2, 0.5, 1.0}
Memory Architecture
Static
Mandate once at init
Memory
Re-inject at every step
Placebo
Length-matched control
LLM Inference Engine
18 Frontier & Open Source LLMs
{ }
At {action, quantity, rationale}
At
Vt
3
Behavioral Evaluation Pipeline
Mandate Drift in Flat Markets
12.7% less drift with memory
Panic Selling in Crashes
12.6% less capital loss with memory
Value Decoupling in Bull Traps
8.8% more rational without memory